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Same Control, Different Verdict

Measuring How Control Wording Changes
Automated AI Compliance Assessments

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Control-Wording Variance Study

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EXECUTIVE SUMMARY

Problem

Organizations increasingly use large language models to assess whether AI systems satisfy security and governance controls such as those in NIST SP 800-53 Revision 5. Those controls are written in English. English admits synonyms, sentence restructuring, and near-equivalent phrasing that a human reviewer can accept as meaning-preserving. If an automated assessor changes its verdict when only the control string changes — evidence held fixed — then control wording is doing security-relevant work. This study measures how often that happens, tests a pre-registered qualifier hypothesis, and asks whether a cataloging-style rewrite reduces wording-driven flips on the controls that moved.

Our objectives were to

- Measure wording variance (X): the share of controls whose 5-way majority verdict flips across meaning-preserving paraphrases of the official statement, on one frozen evidence fixture.
- Test the qualifier hypothesis: whether qualifier-present controls (vague-qualifier + mixed) flip more often than specific controls.
- Test a vocabulary-control rewrite (Y / X2): apply seven editing rules drawn from thesaurus and authority-control practice to the flipped set only, and measure reduction in flip rate without adding, removing, or weakening requirements.

Methodology overview

Part 1 — Control set and paraphrases

Forty official NIST SP 800-53 Rev 5 statements (CSRC spreadsheet Control Text) were selected for applicability to an inspectable ML system. Each control has five variants: official v0 plus four human-reviewed, meaning-preserving paraphrases. SI-4.v1 remains unmarked and is excluded from that control's scored variant set.

Part 2 — Assessment runs and scoring

Every assessment used one frozen mlassure SageMaker-shaped fixture (SHA-256 bded936ec0b5c1c46c091d73e23952e5503e8e2dfe213acdba69711fc50e12de). Assessments were produced by mlassure with local unpushed logging patches. Model claude-sonnet-4-6, temperature 0, three replicas per variant, native 5-way verdicts. Primary X uses majority flip across scored variants; a 3-way OSCAL fail-closed map is reported as sensitivity.

Part 3 — Cataloging rewrite on the flipped set

Seven rules from ANSI/NISO Z39.19, ISO 25964, IFLA GARR, and IFLA ICP were applied only to the ten controls that flipped. After human review (47 yes / 3 unsure excluded), 141 successful reruns measured Y (flip-rate reduction on those ten) and X2 (remaining flips on the original 40-control denominator).

Key findings

- [A]** $X = 10/40 = 25.0\%$. Ten controls change 5-way majority verdict under meaning-preserving paraphrase (AC-6(9), AU-8, CM-2, CM-3, CM-3(1), CM-3(2), CM-8(1), SA-10, SA-4(8), SI-2). Model noise at temperature 0: $32/200 = 16.0\%$ of variants not unanimous. Paraphrase-vs-v0: $22/159 = 13.8\%$.
- [B]** The qualifier hypothesis is not supported. Qualifier-present flipped $2/17 = 11.8\%$; specific flipped $8/23 = 34.8\%$; vague-qualifier alone flipped $0/4$. Specific controls flipped more often in this set.
- [C]** Under 3-way OSCAL fail-closed mapping, $X = 2/40 = 5.0\%$ (AC-6(9) and AU-8 only). Most 5-way flips are not-satisfied $\leftrightarrow$ partially-satisfied and collapse under fail-closed mapping.

[D] Preferred-term rewrite on the flipped ten: $Y = 70\%$ reduction (3/10 still flip: AU-8, CM-2, CM-3). $X2 = 3/40 = 7.5\%$. Rewritten-set noise: $19/47 = 40.4\%$ — higher than Phase 3; reported beside Y, not folded into it.

Recommendations (preview)

- Pin the official catalog statement; do not silently substitute adapted intent.
- Prefer preferred-term / vocabulary-control rewrite over qualifier-stripping; qualifier-stripping was not indicated by this data.
- Persist native 5-way verdicts beside any OSCAL 3-way collapse.
- Report replica noise with X; treat control-text digest as a sibling to the evidence hash.

Conclusion

On this frozen fixture, with one model at temperature 0, control wording moved the 5-way majority on a quarter of a 40-control set. That is enough to treat wording as a security-relevant attribute of an automated assessment. The mechanism is not the one hypothesized: specific controls flipped more than qualifier-present controls. A cataloging-style rewrite reduced majority flips on the treated ten without claiming that library science solved compliance, and without lobbying for catalog change. Results are bounded by one fixture, one model, forty controls, and unpushed assessment patches.

PROJECT INFORMATION

Title	Same Control, Different Verdict: Measuring How Control Wording Changes Automated AI Compliance Assessments
Author	Jose Ruiz-Vazquez
Study period	August 2026 (assessment runs); PDF dated September 2026
Status	Private study draft / study archive — not published
Assessment tool	mlassure (github.com/joseruiz1571/mlassure), with local unpushed logging patches
Archive	Private study archive; local patches and run logs are not published with this draft
Model	claude-sonnet-4-6, temperature 0, three replicas per variant
Control source	NIST SP 800-53 Rev 5 — CSRC spreadsheet Control Text (official v0)
Evidence	One frozen SageMaker-shaped mlassure fixture; SHA-256 bded936ec0b5c1c46c091d73e23952e5503e8e2dfe213acdba69711fc50e12de

No external funding is claimed for this study. This document does not constitute a product endorsement, a NIST recommendation, or a published research article. Official NIST control text is a U.S. government work.

METHODOLOGY

Part 1 — Control set and paraphrases

Objective. Construct a human-reviewed set of meaning-preserving paraphrases of official NIST SP 800-53 Rev 5 statements so that wording can be varied while requirements stay fixed.

Data. Forty controls were selected because they apply to a machine-learning system an automated assessor can inspect (families AC, AU, CM, RA, SA, SI, SC, CA). Deterministic and attestation-only controls were excluded: their verdicts are not produced by reading intent, so wording cannot change them. Official v0 wording is the CSRC spreadsheet Control Text, not adapted fixture intents and not reconstructed OSCAL parameter inserts. Assignment and Selection blocks are copied verbatim.

Each control has five versions (v0 plus four paraphrases). Paraphrases change vocabulary and sentence structure only; they do not add SHALL/MUST, numbers, or roles absent from v0. Human review marked 199 rows meaning-preserved yes. SI-4.v1 was rewritten after review and remains unmarked; it is excluded from SI-4's scored variant set for X, for vocab-pattern flip tables, and for paraphrase-vs-v0. SI-4 is therefore scored on four variants (v0, v2, v3, v4). Vocab tags on v0: 4 vague-qualifier, 23 specific, 13 mixed. The hypothesis cut is qualifier-present (vague-qualifier + mixed = 17) versus specific (23).

Part 2 — Assessment runs and scoring

Objective. Hold evidence fixed and measure how often majority verdicts change when only the control string changes.

Frozen evidence. One mlassure SageMaker-shaped fixture (churn-predictor-v1 style), SHA-256 bded936ec0b5c1c46c091d73e23952e5503e8e2dfe213acdba69711fc50e12de. Chosen for mixed satisfied and not-satisfied conditions so wording has room to flip. It is not an mltrack bundle; mlassure cannot consume mltrack bundles. The hash was re-verified at scoring and is unchanged through Phase 5.

Assessor. Assessments were produced by mlassure with local logging patches that remain unpushed. Control wording is supplied as `ControlItem.intent`. Verdicts are five-way: satisfied, partially-satisfied, not-satisfied, not-applicable, insufficient-evidence. Model: claude-sonnet-4-6. Temperature: 0 on every successful replica.

Replicas and census. Each variant was assessed three times. Scoring uses the last successful jsonl record per variant_id + replica; failed credit-crash rows are ignored. Result: 600 unique successful replicas (200 variants × 3). Census: 454 not-satisfied, 132 partially-satisfied, 10 satisfied, 4 insufficient-evidence (0 not-applicable).

Scoring rules. Majority is the unique status with count ≥ 2 among three replicas; a 1-1-1 triple is labeled split. A control flips when its scored variant majority verdicts are not all identical. Primary X = flipped controls / 40 (SI-4 contributes with four scored variants). Paraphrase-vs-v0 counts meaning-preserved paraphrases whose majority differs from that control's v0 majority (denominator 159 after excluding SI-4.v1).

3-way sensitivity map (OSCAL fail-closed), applied per replica then re-majored: satisfied → satisfied; not-satisfied → not-satisfied; insufficient-evidence → undetermined; partially-satisfied → not-satisfied; not-applicable → not-satisfied. This map is a labeled sensitivity, not the primary outcome.

Part 3 — Cataloging rewrite on the flipped set

Objective. Ask whether vocabulary-control editing reduces flip rate on the controls that already moved — without adding, removing, or weakening requirements, and without injecting fixture artifact names into control text.

Seven editing rules (R1-R7) were drawn from sources actually retrieved: ANSI/NISO Z39.19-2005

(R2010), ISO 25964-1:2011, IFLA GARR (2nd ed., 2001), and IFLA ICP (2016). Rules address preferred terms, named count-noun evidence objects already present in the control, factoring of compound actions, scoped class terms, closed OR-selections, explicit associative links, and discrete authorized headings for lettered clauses. Assignment and Selection blocks are copied verbatim.

Scope. Rules were applied only to the ten flipped controls, not to all forty. Human review of 50 rewritten rows: 47 yes, 3 unsure (AC-6(9).r4, AU-8.r4, CM-2.r4). Unsure rows are excluded. Remaining 47 variants × 3 replicas = 141 successful assessments (temperature 0, same model, same frozen hash).

$Y = (10 - n_flipped_after) / 10$, the reduction in flip rate on the treated ten. Before rewrite, all ten flipped. $X2 = n_flipped_after / 40$, counting the 30 untreated controls as still non-flipping.

FINDINGS

Primary wording variance (X)

$X = 10/40 = 25.0\%$ under 5-way majority scoring (SI-4 scored on four variants). Dropping SI-4 entirely yields $X_{39} = 10/39 = 25.6\%$; SI-4 itself does not flip (four $\times$ not-satisfied). The ten flipped controls are: AC-6(9), AU-8, CM-2, CM-3, CM-3(1), CM-3(2), CM-8(1), SA-10, SA-4(8), SI-2.

Paraphrase-versus-v0: 22 of 159 meaning-preserved paraphrases (13.8%) differ from that control's v0 majority. All 22 sit inside the ten flipped controls.

Replica noise

Even at temperature 0, $32/200 = 16.0\%$ of variants were not unanimous across three replicas (31 two-one majorities; 1 three-way split on CM-2.v4). Noise is reported with X because majority scoring does not erase replica disagreement.

Qualifier hypothesis — not supported

The study expected qualifier-present controls to flip more often than specific controls. The completed table goes the other way. This result is reported honestly; the hypothesis is not supported.

Cut	n controls	n flipped	Flip rate
vague-qualifier	4	0	0/4 = 0.0%
mixed	13	2	2/13 = 15.4%
specific	23	8	8/23 = 34.8%
qualifier-present (vague + mixed)	17	2	2/17 = 11.8%

Vague-qualifier alone (AC-6, RA-7, SA-8(32), SI-12) flipped 0/4. Mixed flips were CM-2 and SA-4(8). Specific controls flipped more often than qualifier-present controls in this 40-control set. This paper does not claim the hypothesis was confirmed, qualified, or directionally right — and it does not claim that vague controls are the problem.

Three-way OSCAL sensitivity

After fail-closed mapping, 3-way $X = 2/40 = 5.0\%$. Only AC-6(9) and AU-8 move satisfied $\leftrightarrow$ not-satisfied. Most 5-way flips are not-satisfied $\leftrightarrow$ partially-satisfied (plus one split); those collapse under the map. 3-way noise is $9/200 = 4.5\%$; 3-way paraphrase-vs-v0 is $6/159 = 3.8\%$. If a downstream program treats partially-satisfied as “not met,” eight of the ten 5-way flips are operationally silent. Primary X keeps the five-way enum the tool emits.

Rewrite results (Y and X2)

Before rewrite, all ten treated controls flipped. After rewrite and exclusion of three unsure paraphrases, three still flip: AU-8, CM-2, and CM-3. Seven no longer flip: AC-6(9), CM-3(1), CM-3(2), CM-8(1), SA-10, SA-4(8), and SI-2.

Metric	Value
Y (flip-rate reduction on treated 10)	70% (10/10 $\rightarrow$ 3/10)
X2 (remaining flips / original 40)	3/40 = 7.5%
Still flipping after rewrite	AU-8, CM-2, CM-3
Rewritten-set noise	19/47 = 40.4%
Phase 5 successful runs	141 (47 variants $\times$ 3)

Phase 5 replica census (141): 90 not-satisfied, 40 partially-satisfied, 8 satisfied, 3 insufficient-evidence. Rewritten-set noise (40.4%) is higher than Phase 3 noise (16.0%). Y is a

reduction in majority-verdict wording variance; the noise rate is reported beside it, not folded into it. Y is not “the model became more stable.”

LIMITATIONS AND FUTURE WORK

Binding limitations

- One frozen evidence fixture. Results are not a claim about other inventories, production systems, or evidence shapes.
- One model (claude-sonnet-4-6) at temperature 0. Temperature 0 does not imply determinism; 16.0% Phase 3 noise is the measurement.
- Forty controls. Results are not a ranking of NIST families and are not a catalog rewrite proposal.
- SI-4.v1 hole: unmarked and excluded from SI-4's scored variant set. The four scored SI-4 variants already agree (no flip), so the hole does not change X.
- Three rewritten paraphrases marked unsure and excluded (AC-6(9).r4, AU-8.r4, CM-2.r4), so those three controls are scored on four rewritten variants rather than five.
- mlassure logging patches are local and unpushed. Reproducing live assessment requires a patched clone and an API key; table rebuild from frozen logs does not.

This study does not lobby NIST, does not claim library science solved compliance, and does not claim that vague controls are the problem. The qualifier hypothesis was not supported; recommendations follow the data.

Future work licensed by this data

- Second model on the same frozen fixture and control set, to separate assessor-specific wording sensitivity from fixture-specific effects.
- Second evidence fixture with a different mix of satisfied / not-satisfied conditions, holding control variants fixed.
- Persist a control-text digest in the assessment artifact as a sibling to the evidence hash, so wording provenance is auditable beside evidence provenance.
- Residual analysis on AU-8, CM-2, and CM-3 — the three controls that still flip after preferred-term rewrite — without expanding scope to untreated controls.

Out of scope: employee-frame narratives, NIST lobbying, catalog-wide rewrite claims, and any assertion that qualifier-stripping is the remedy. Those are refused because the data do not license them.

RECOMMENDATIONS

These recommendations are scoped to automated LLM assessment pipelines that read control text. They are operational hygiene for wording provenance and scoring transparency, not a call to rewrite the NIST catalog.

1. Pin the official statement

Treat the official catalog Control Text as the pinned input. Do not silently substitute a vendor's adapted intent for the catalog statement when measuring or reporting compliance. If adapted wording is used, record it as a distinct artifact with its own digest.

2. Prefer preferred-term rewrite over qualifier-stripping

Where control authors or assessment designers edit wording for machine readability, prefer vocabulary-control practice: one preferred term per required action or object; named count-noun artifacts already present in the control; factored compound actions; closed selections; explicit associative links. Do not assume that stripping vague qualifiers will stabilize automated verdicts — it was not indicated here, because qualifier-tagged controls were not the ones that flipped.

3. Persist 5-way verdicts beside OSCAL

Keep the native five-way enum the assessor emits, and map to OSCAL's fail-closed three-way only as an explicit downstream view. Collapsing partially-satisfied before measuring wording variance hides the majority of flips observed in this study (3-way $X = 2/40$ vs 5-way $X = 10/40$).

4. Report noise with X

Publish replica disagreement beside the flip rate. Majority scoring does not erase temperature-0 noise (16.0% in Phase 3; 40.4% on the rewritten set). Readers need both numbers to interpret stability claims.

5. Treat control-text digest as sibling to evidence hash

Assessment artifacts should record a digest of the exact control string assessed, alongside the evidence hash. Wording provenance and evidence provenance are parallel integrity concerns when an LLM is the reader.

End of report. Private study draft — Jose Ruiz-Vazquez — August-September 2026. Not published. Frozen evidence SHA-256 bded936ec0b5c1c46c091d73e23952e5503e8e2dfe213acdba69711fc50e12de.